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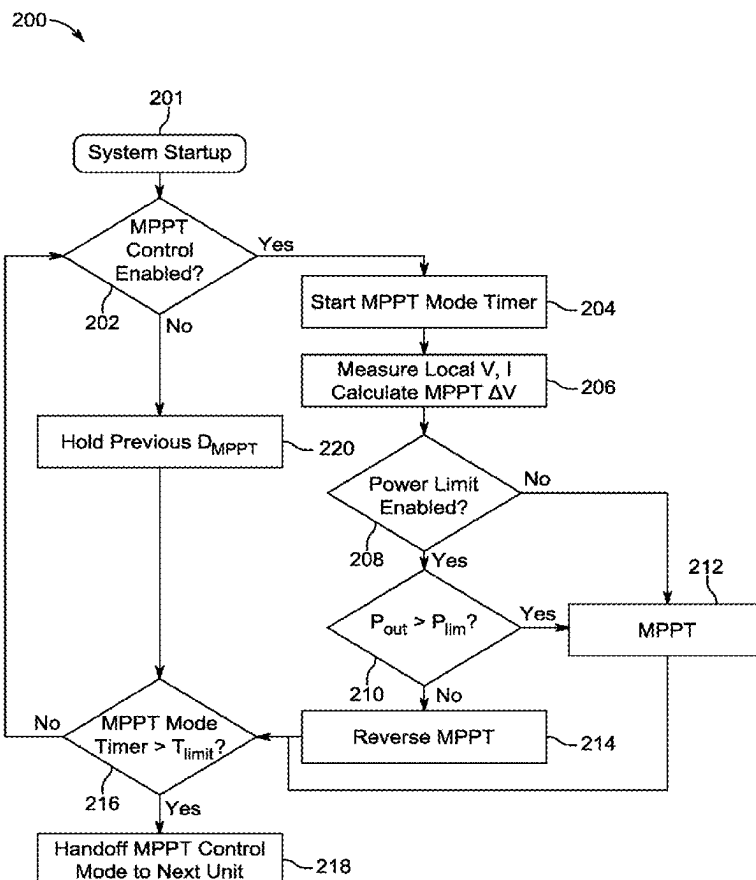
(19) **United States**(12) **Patent Application Publication** (10) **Pub. No.: US 2025/0278107 A1**
(43) **Pub. Date: Sep. 4, 2025**(54) **SYSTEMS AND METHODS FOR
PHOTOVOLTAIC DIRECT CURRENT (DC)
BUS CONTROL**(52) **U.S. Cl.**
CPC *G05F 1/67* (2013.01); *H02J 3/381*
(2013.01); *H02J 7/35* (2013.01); *H02J*
2207/20 (2020.01); *H02J 2300/26* (2020.01)(71) Applicant: **Nextracker LLC**, Fremont, CA (US)(72) Inventors: **Yang LIU**, Mountain View, CA (US);
Alexander W. AU, Oakland, CA (US);
Fei GU, Fremont, CA (US)(57) **ABSTRACT**(21) Appl. No.: **19/214,243**(22) Filed: **May 21, 2025****Related U.S. Application Data**

- (63) Continuation of application No. 18/409,086, filed on Jan. 10, 2024, now Pat. No. 12,339,692, which is a continuation of application No. 17/601,921, filed on Oct. 6, 2021, filed as application No. PCT/US2020/031303 on May 4, 2020, now Pat. No. 11,907,000.
- (60) Provisional application No. 62/842,798, filed on May 3, 2019.

Publication Classification

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G05F 1/67 (2006.01)
H02J 3/38 (2006.01)
H02J 7/35 (2006.01)

Solar power systems and methods utilize DC power transmission and centralized power inversion. The solar power systems include a photovoltaic (PV) bus system and a fixed bus system. The PV system utilizes a control mode handoff control method, which includes determining that a local maximum power point tracking (MPPT) control is enabled; in response to determining that the local MPPT control is enabled, starting an MPPT mode timer; performing local MPPT; determining that the MPPT mode timer is greater than a predetermined period; and, in response to determining that the MPPT mode timer is greater than a predetermined period, handing off MPPT control to the next MPPT controller. The distributed MPPT control method may include sequential MPPT control, adaptive ΔV MPPT control, and/or power limiting control. The fixed bus system includes PV string-level MPPT controllers and a fixed DC input central inverter or multiple fixed DC modular inverters.



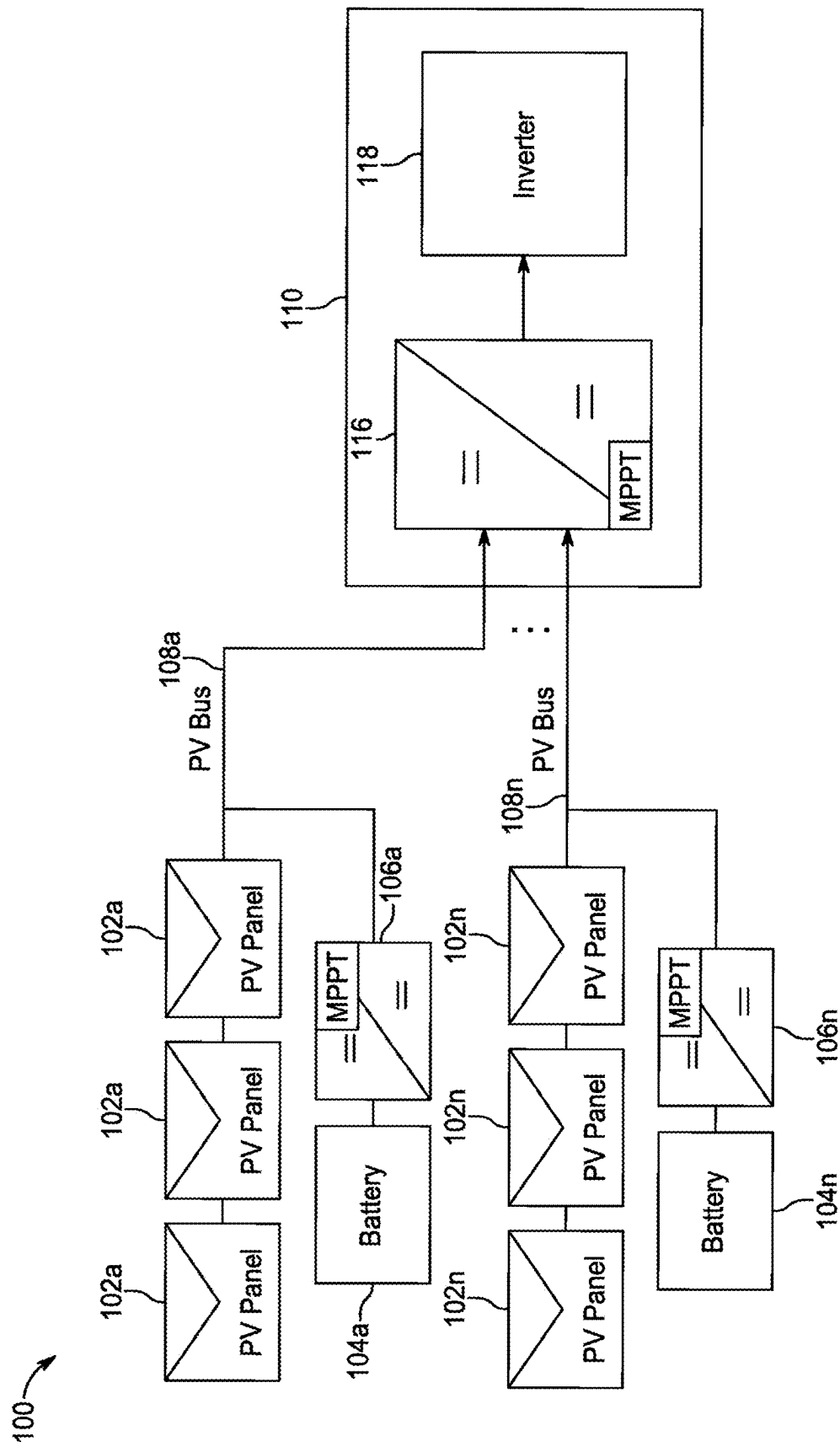


FIG. 1

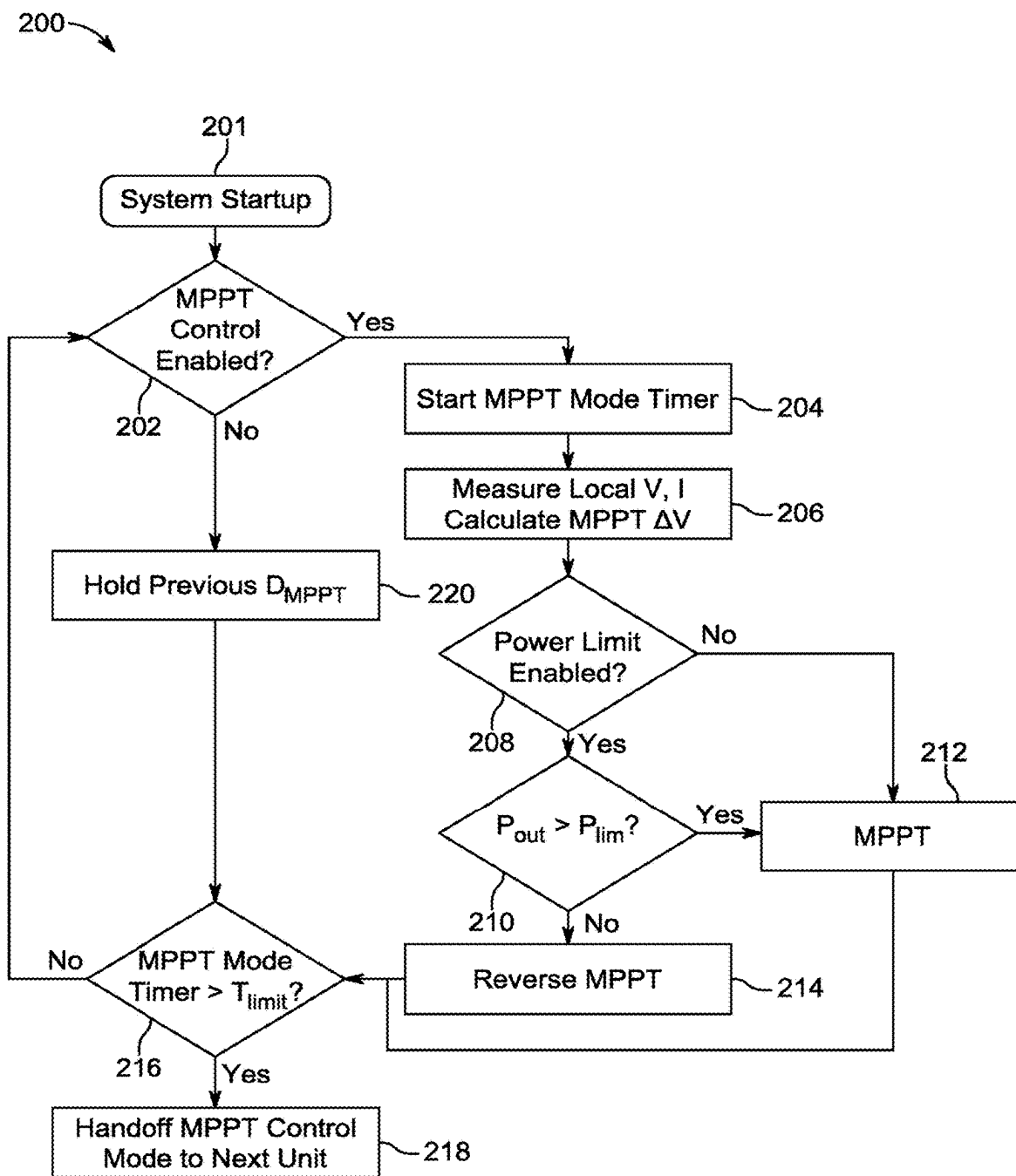


FIG. 2

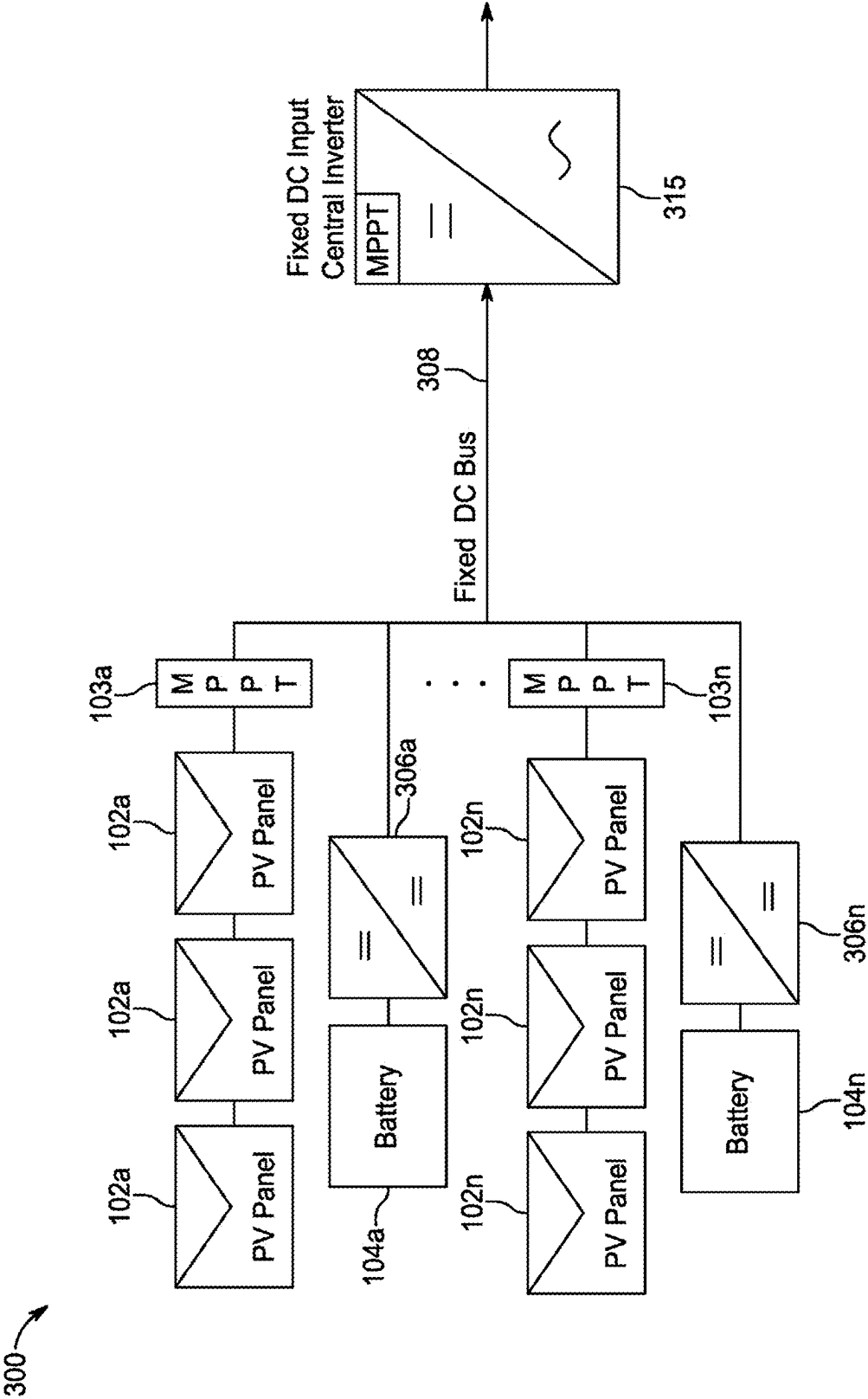


FIG. 3

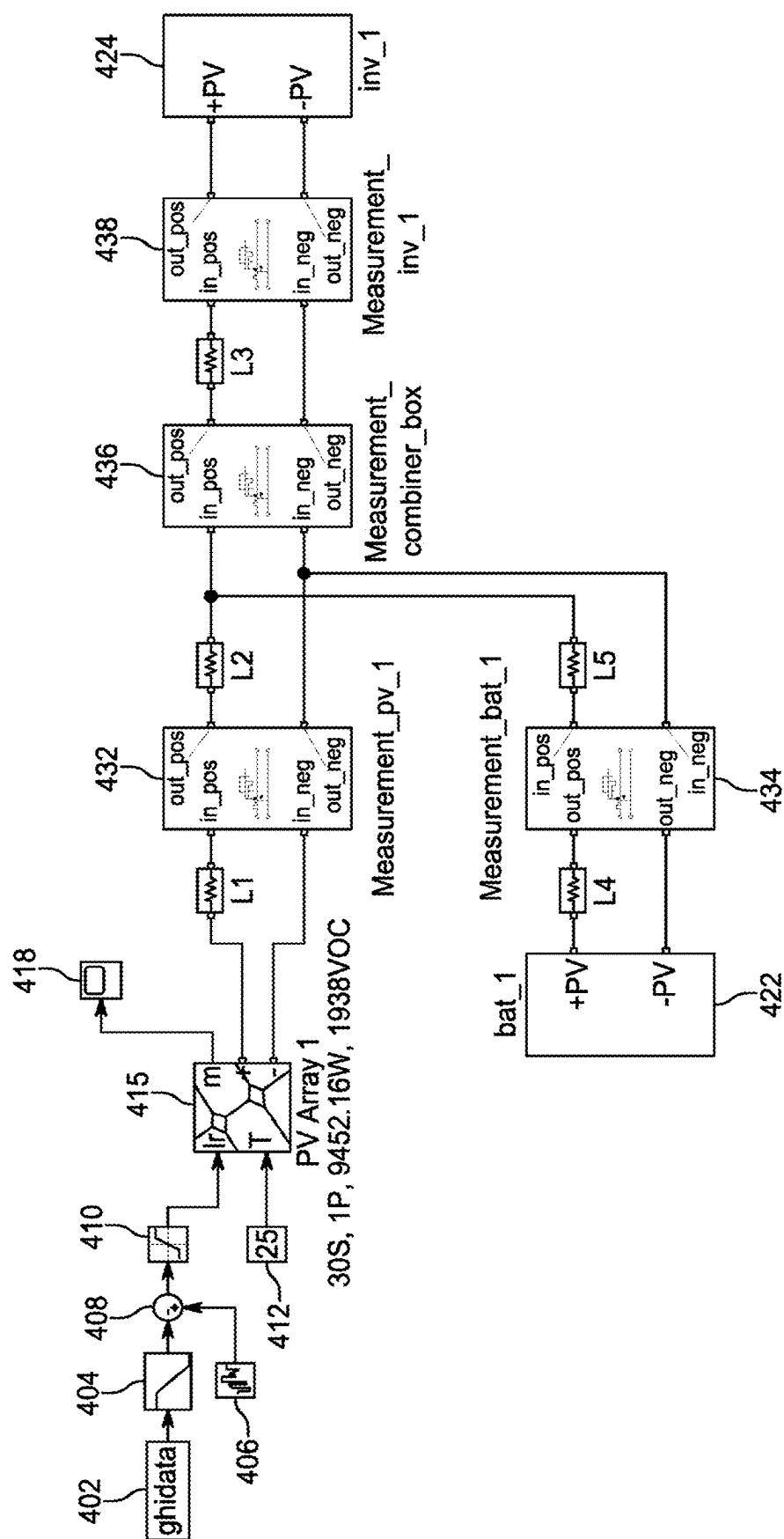


FIG. 4

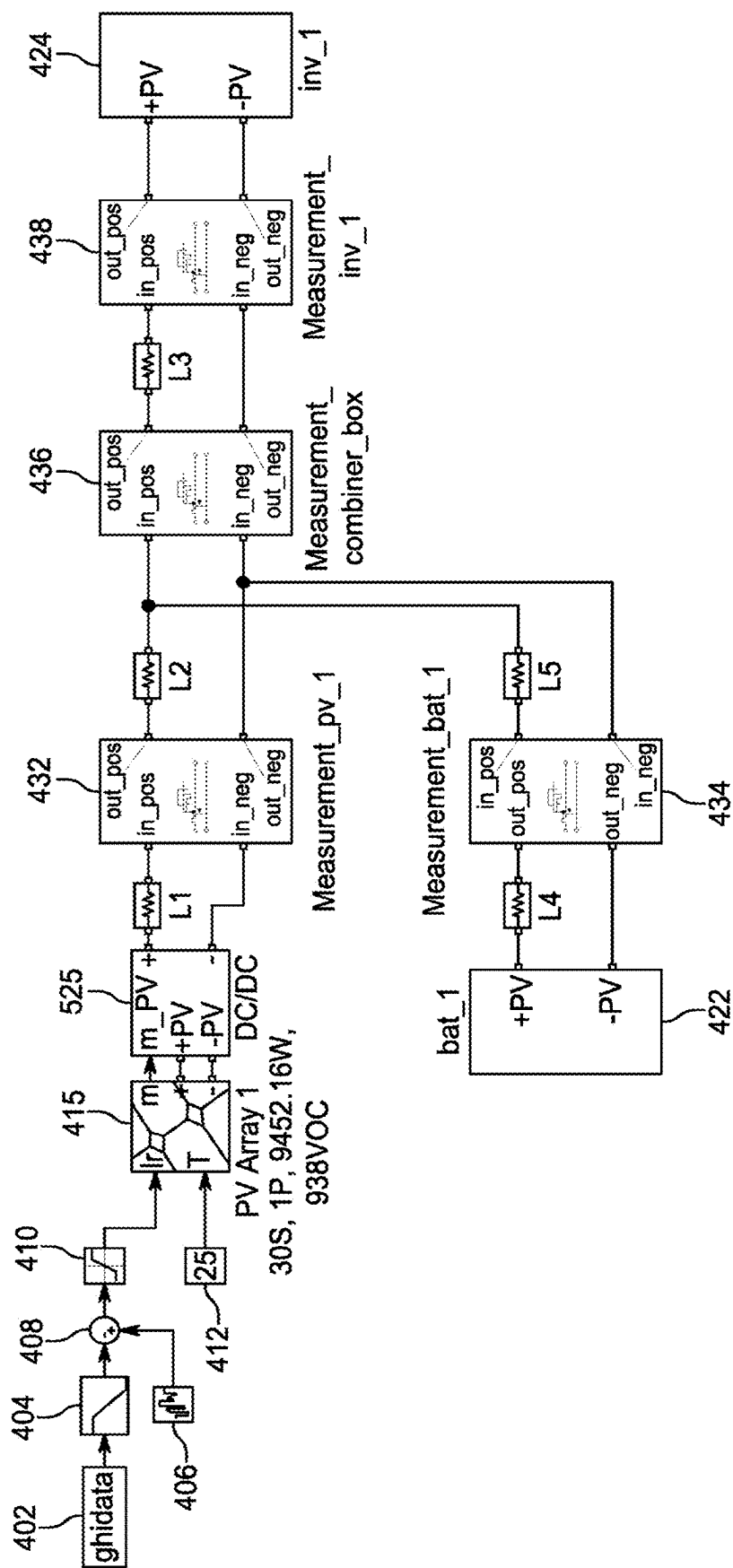


FIG. 5

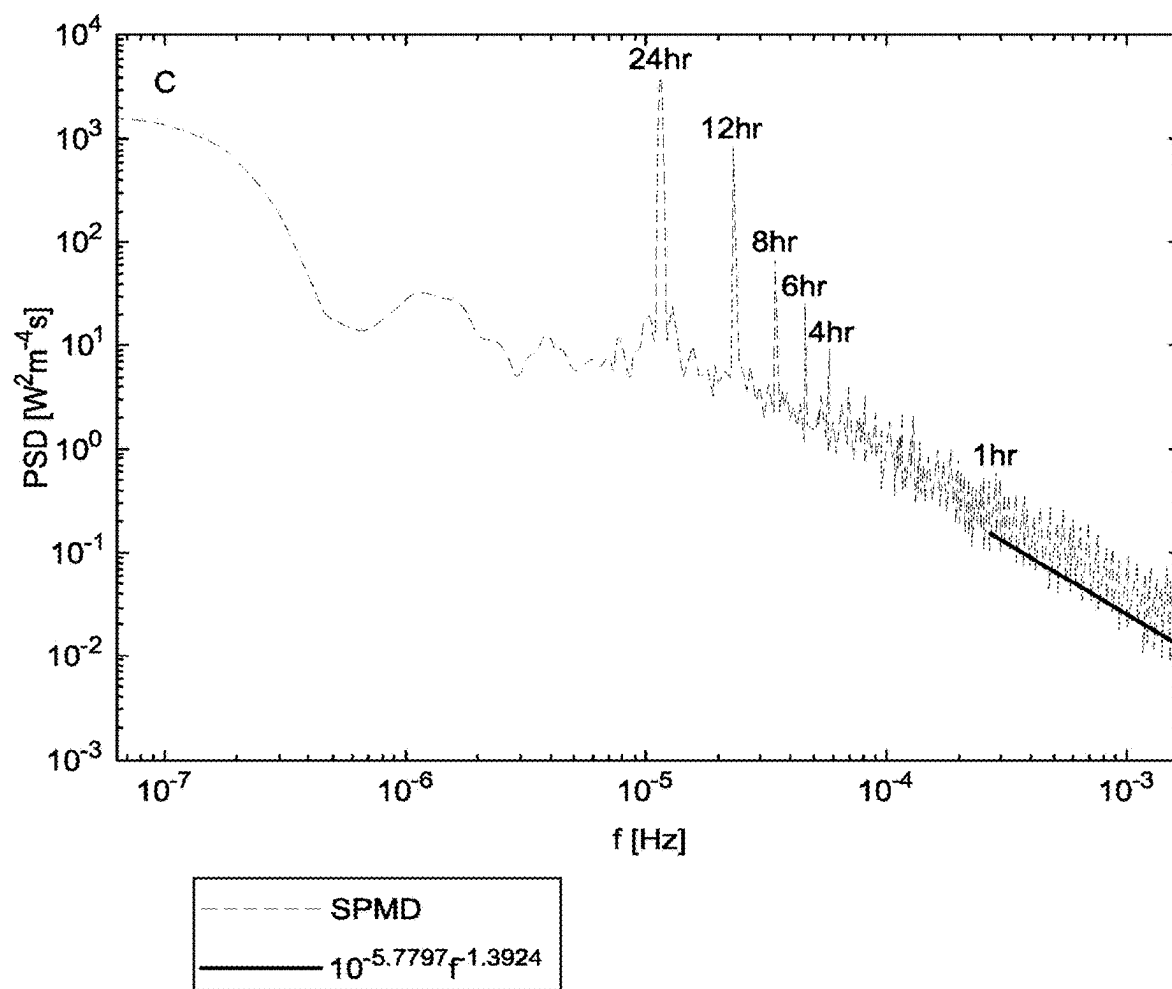


FIG. 6

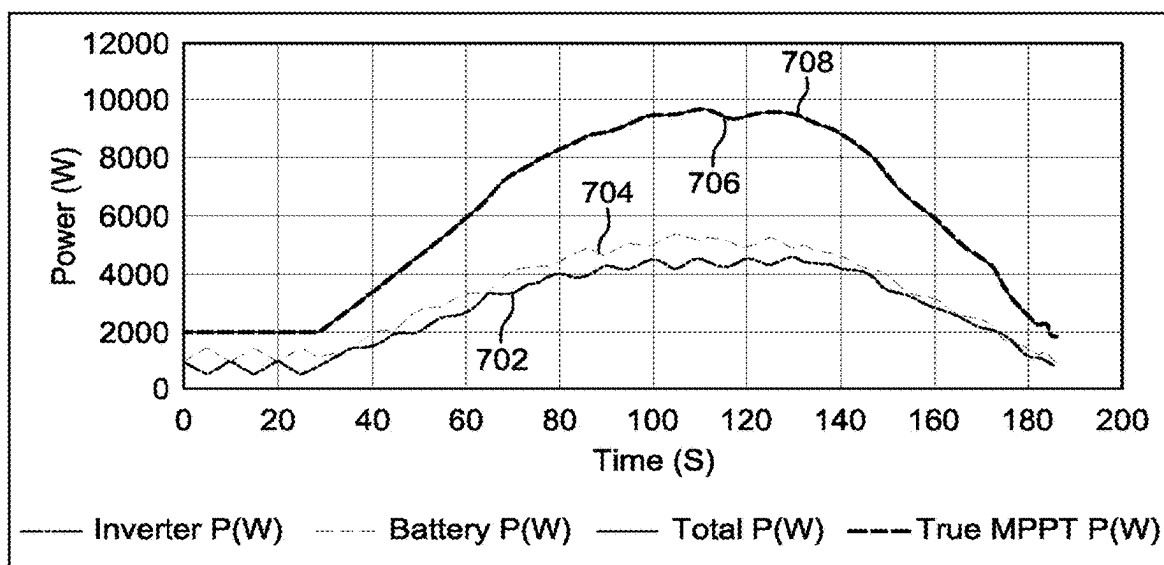


FIG. 7

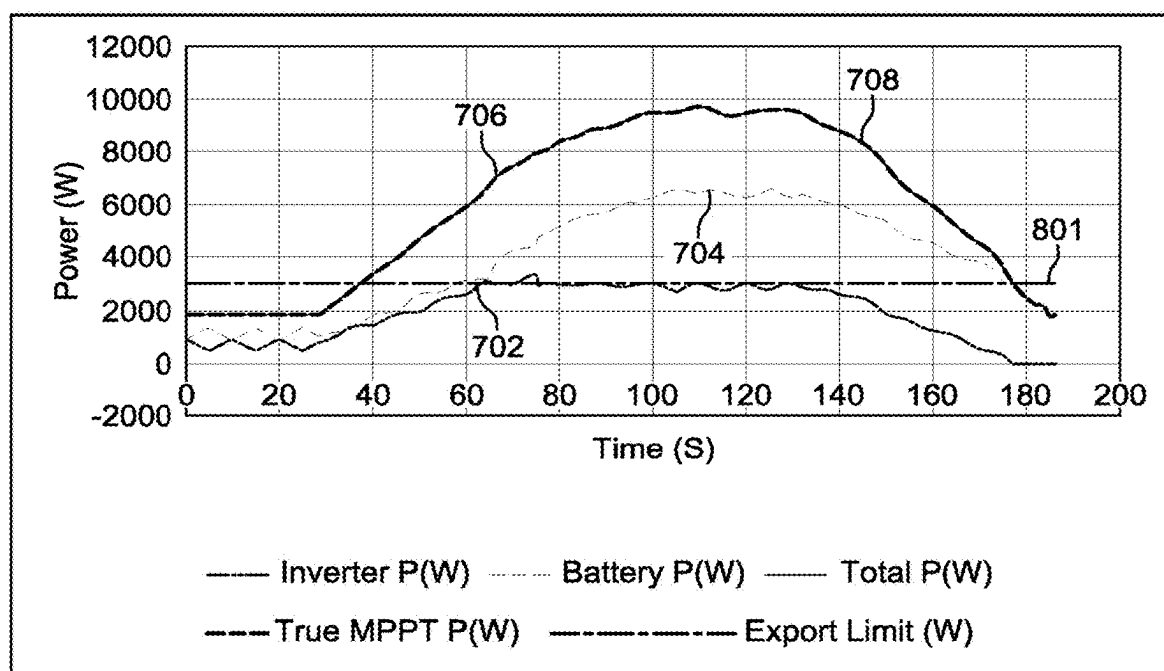


FIG. 8

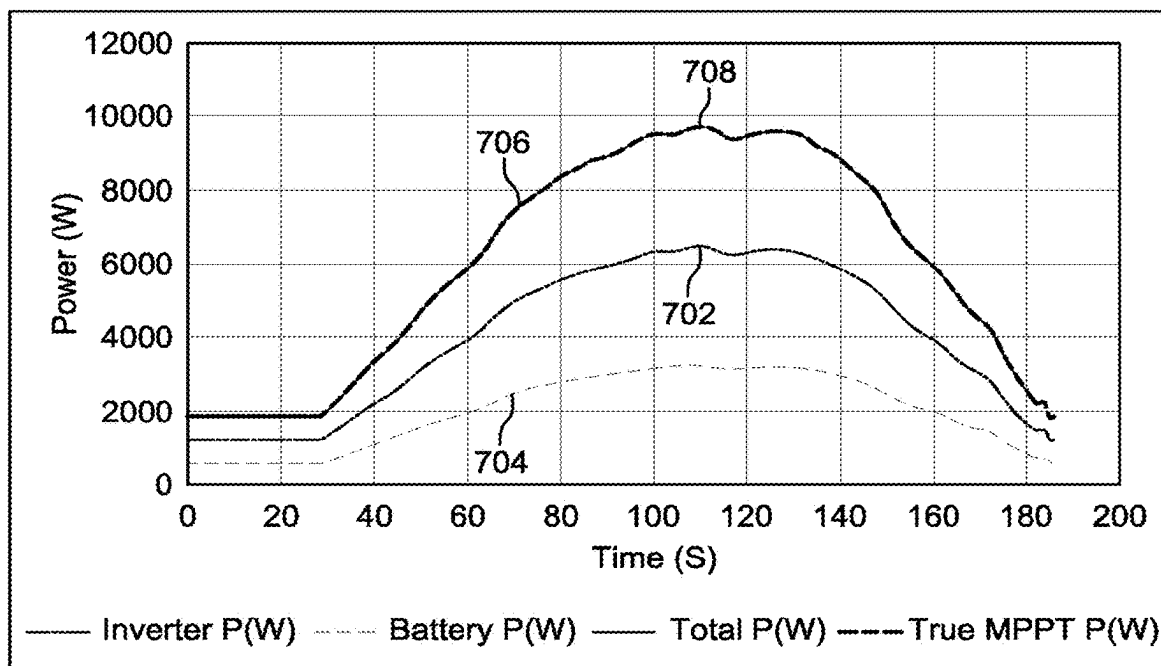


FIG. 9

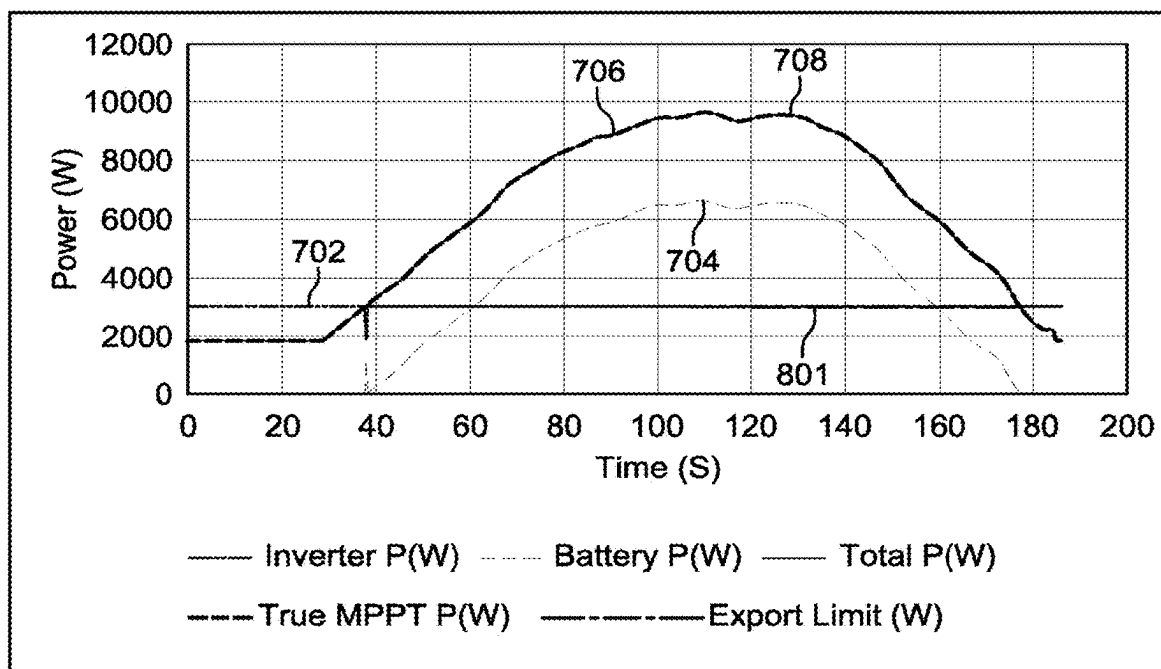


FIG. 10

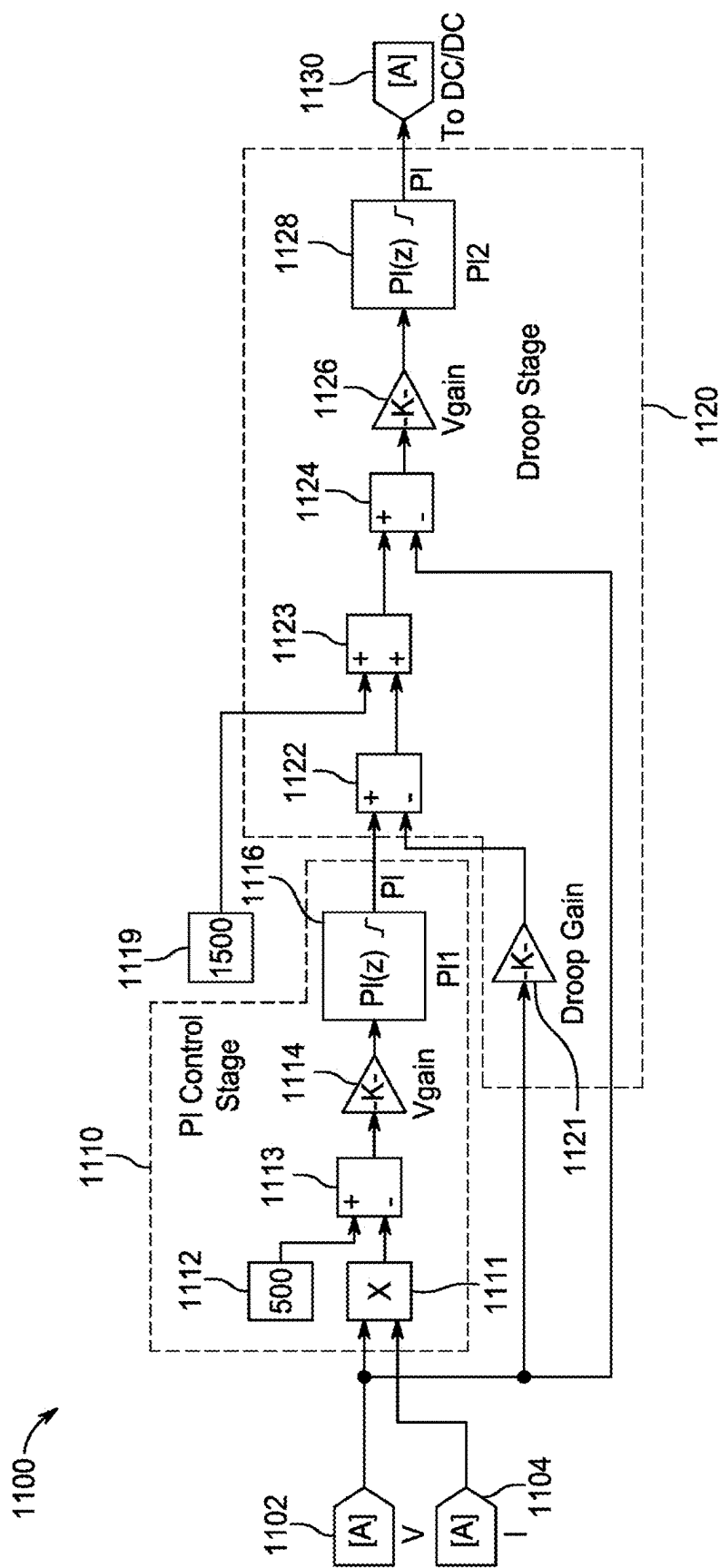


FIG. 11

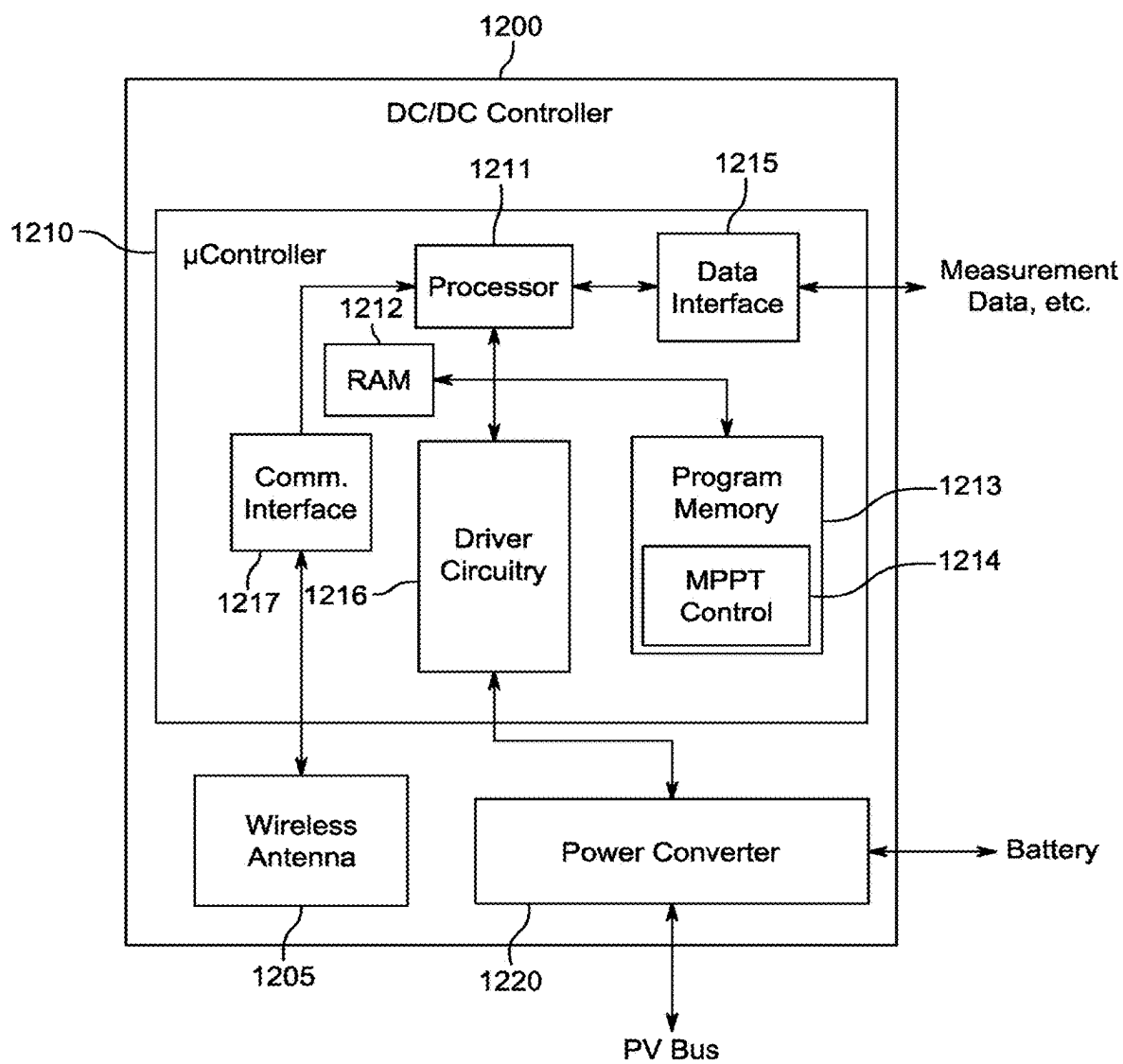


FIG. 12

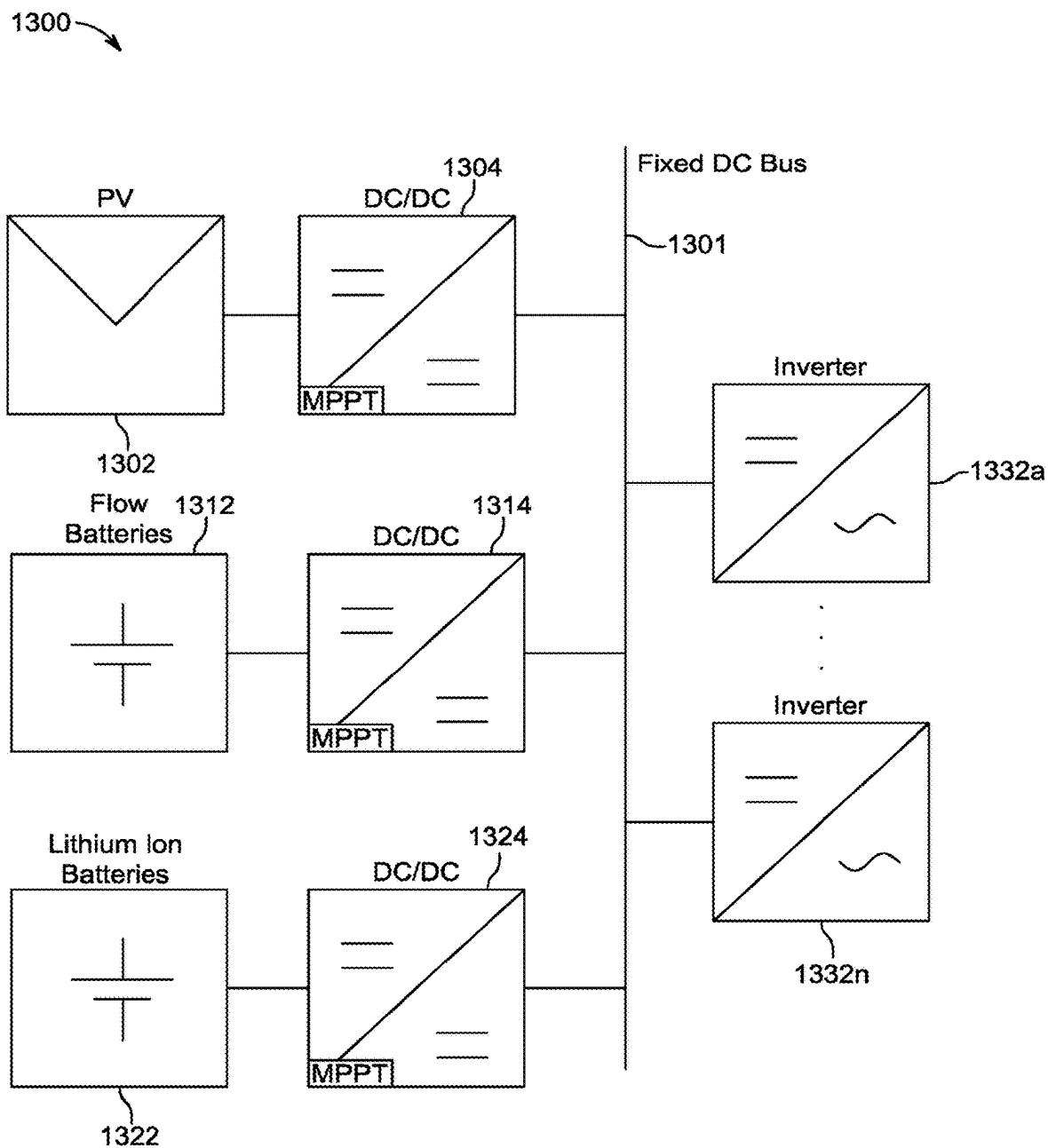


FIG. 13

SYSTEMS AND METHODS FOR PHOTOVOLTAIC DIRECT CURRENT (DC) BUS CONTROL

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/409,086, filed Jan. 10, 2024, which is a continuation of U.S. patent application Ser. No. 17/601,921, filed Oct. 6, 2021, now U.S. Pat. No. 11,907,000, which is a 371 US National Stage of PCT Application No. PCT/US2020/031303 filed May 4, 2020, which the benefit of U.S. Provisional Application No. 62/842,798, filed May 3, 2019, the entire contents of each of which are incorporated herein by reference.

FIELD

[0002] This disclosure is generally directed to solar power generating systems. More particularly, this disclosure is directed to solar power systems and methods utilizing DC power transmission and centralized power inversion.

BACKGROUND

[0003] Solar and wind energy are increasingly important renewable, non-polluting energy sources for consumers and businesses throughout the world. For solar energy, photovoltaic (PV) panels arranged in an array or string typically provide the means to convert solar energy into electrical energy. In operating PV arrays, maximum power point tracking (MPPT) is generally used to automatically determine a voltage or current at which the PV array should operate to generate a maximum power output for a particular temperature and solar irradiance. Although MPPT allows for the generation of maximum output power, the transmission and storage of the power generated by the PV arrays may be inefficient and costly.

SUMMARY

[0004] The techniques of this disclosure generally relate to PV bus and fixed DC bus power systems, and methods of controlling the PV bus and fixed DC power systems. Among other advantages, the techniques of this disclosure provide redundancy among a network of DC/DC controllers in case a subset of the DC/DC controllers fails or is unable to perform MPPT functions.

[0005] One general aspect includes a method for controlling a PV bus power system. The method includes determining, at a current DC/DC controller, that local maximum power point tracking (MPPT) control is enabled; in response to determining that local MPPT control is enabled, starting an MPPT mode timer; performing MPPT; determining that the MPPT mode timer is greater than a predetermined period; and, in response to determining that the MPPT mode timer is greater than a predetermined period, handing off MPPT control mode to the next DC/DC controller. Other aspects include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

[0006] Implementations may include one or more of the following features. The method may include determining that the local MPPT control is not enabled; and in response to determining that the local MPPT control is not enabled,

holding the previous duty cycle at the current MPPT controller's last known MPPT setpoint. The method may include measuring local voltage and current; and determining maximum power point voltage seeking direction ΔV based on the measured local power perturbations. The method may include determining that a power limit is enabled; and in response to determining that the power limit is enabled, determining the output power based on the measured local voltage and current, and determining that the output power is greater than a power threshold; and in response to determining that the output power is greater than the power threshold, performing MPPT in the direction of negative ΔV . The method may include determining that the output power is not greater than a power threshold; and, in response to determining that the output power is not greater than the power threshold, performing MPPT in the direction of positive ΔV , also referred to herein as reverse MPPT.

[0007] The method may include determining that a power limit is not enabled; and, in response to determining that the power limit is not enabled, performing MPPT. The method may include determining that the MPPT mode timer is not greater than the predetermined period; in response to determining that the MPPT mode timer is not greater than the predetermined period, further determining that the MPPT control is enabled; and, in response to further determining that the MPPT control is enabled, starting the MPPT mode timer. The method may include determining that the MPPT mode timer is not greater than the predetermined period; in response to determining that the MPPT mode timer is not greater than the predetermined period, further determining that the MPPT control is not enabled; and, in response to further determining that the MPPT control is not enabled, further holding the previous duty cycle at the current MPPT controller's last known MPPT setpoint.

[0008] The method may include in response to determining that the MPPT mode timer is greater than a predetermined period, allowing a voltage perturbation for a distributed energy resource before handing off MPPT control mode to the next MPPT controller. The method may include scaling the magnitude of the voltage perturbation. The method may include scaling the magnitude of the voltage perturbation based on a local device metric. The local device metric may be an output power or a battery state of charge. Implementations of the described techniques may include hardware, a method or process, or computer software on a computer-accessible medium.

[0009] Another general aspect features a PV bus system. The PV bus system includes a central inverter unit, PV buses coupled to the central inverter unit, PV strings coupled to the central inverter unit via the PV buses, respectively, batteries coupled to the PV buses, respectively, and DC/DC controllers coupled between the batteries and the PV buses, respectively. Each of the DC/DC controllers includes a processor and a memory. The memory has stored thereon instructions, which, when executed by the processor, causes the DC/DC controller to determine that a maximum power point tracking (MPPT) control is enabled; in response to determining that the local MPPT control is enabled, start an MPPT mode timer; perform MPPT; determine that the MPPT mode timer is greater than a predetermined period; and in response to determining that the MPPT mode timer is greater than a predetermined period, hand off MPPT control mode to a next DC/DC controller.

[0010] Implementations of the PV bus system may include one or more of the following features. Each of the DC/DC controllers may include a MPPT controller and a DC/DC converter. Each of the DC/DC controllers may include a proportional-integral-derivative (PID) control stage and a droop stage coupled to the PID control stage. Each of the PV strings may include at least one PV panel. The instructions, when executed by the processor, may further cause the DC/DC controller to determine that the MPPT control is not enabled; and, in response to determining that the MPPT control is not enabled, hold the previous duty cycle at the current MPPT controller's last known MPPT setpoint. The instructions, when executed by the processor, may further cause the DC/DC controller to measure local voltage and current; determine MPPT ΔV based on the measured local voltage; determine that a power limit is enabled; and, in response to determining that the power limit is enabled, determine the output power based on the measured local voltage and current; and determine that the output power is greater than a power threshold; and, in response to determining that the output power is greater than the power threshold, perform MPPT. The instructions, when executed by the processor, may further cause the DC/DC controller to determine that the output power is not greater than a power threshold; and, in response to determining that the output power is not greater than the power threshold, perform reverse MPPT.

[0011] Still another general aspect features a fixed DC bus system. The fixed DC bus system includes modular inverters; a fixed DC bus coupled to the modular inverters; PV strings coupled to the modular inverters via the fixed DC bus; batteries coupled to the fixed DC bus; and DC/DC controllers coupled between the batteries and the fixed DC bus. Each PV string includes an MPPT controller. The MPPT controller includes a PID control stage and a droop stage having a droop gain.

[0012] Implementations of the fixed DC bus system may include one or more of the following features. The PID control stage may include a first voltage gain block and a first proportional-integral (PI) controller coupled to the output of the first voltage gain block. The droop stage may include a droop gain block that applies a droop gain to an input voltage to obtain a droop gain voltage, and a first subtraction block that is coupled to the output of the droop gain block and that subtracts the droop gain voltage from the output of the first PI controller to obtain an adjusted PI controller output. The droop stage may include an addition block that is coupled to the output of the first subtraction block and that adds the adjusted PI controller output to a reference power to obtain a further adjusted PI controller output. The droop stage may include a second subtraction block that is coupled to the addition block and that subtracts the input voltage from the further adjusted PI controller output. The droop stage may include a second voltage gain block that is coupled to the output of the second subtraction block, and a second PI controller that is coupled to the output of the second voltage gain block.

[0013] The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

[0014] Various aspects of the present disclosure are described herein below with reference to the drawings, which are incorporated in and constitute a part of this specification, wherein:

[0015] FIG. 1 is a schematic diagram illustrating a PV bus system;

[0016] FIG. 2 is a flowchart illustrating MPPT control mode handoff for PV bus systems;

[0017] FIG. 3 is a schematic diagram illustrating a fixed DC bus system architecture according to one aspect of this disclosure;

[0018] FIG. 4 is a schematic diagram illustrating a PV bus model;

[0019] FIG. 5 is a schematic diagram illustrating a fixed DC bus model;

[0020] FIG. 6 is a graphical diagram illustrating solar variability spectra, which may be used to determine the rate at which MPPT control may be handed off between two MPPT units of a network of MPPT units;

[0021] FIG. 7 is a graphical diagram illustrating PV bus MPPT performance;

[0022] FIG. 8 is a graphical diagram illustrating PV bus MPPT and power export limiting performance;

[0023] FIG. 9 is a graphical diagram illustrating fixed DC bus MPPT and voltage droop power sharing performance;

[0024] FIG. 10 is a graphical diagram illustrating fixed DC bus power export limiting with MPPT performance;

[0025] FIG. 11 is a schematic diagram illustrating a hybrid power limiting droop controller;

[0026] FIG. 12 is a schematic diagram illustrating an example of a DC/DC controller according to this disclosure; and

[0027] FIG. 13 is a schematic diagram illustrating a fixed DC bus system architecture according to another aspect of this disclosure.

DETAILED DESCRIPTION

[0028] The fixed DC bus system architecture (e.g., the fixed DC bus system **300** illustrated in FIG. 3) of this disclosure includes a fixed DC bus that serves as a standardized power interconnection for photovoltaic (PV) generation and storage devices (e.g., PV arrays, batteries, controllers, drivers, alternating current (AC) systems, etc.). In aspects, common fixed bus systems may address some of the issues associated with PV- and battery-bus systems and accelerate the deployment and improve the cost savings of complete photovoltaic/energy storage systems.

[0029] To interface PV arrays to a fixed DC bus, distributed maximum power point tracking (MPPT) controllers at the string, sub-string, or module level are needed to maximize PV power generation. The increased cost of distributed MPPT controllers may be sufficiently offset by cost savings in wiring and improvements in total energy yield.

[0030] Central inverter PV bus systems (e.g., the PV bus system **100** illustrated in FIG. 1), in which a central inverter may drive the MPPT voltage of all PV strings connected to the central inverter, may mirror features inherent in fixed DC bus systems, including simultaneous MPPT tracking, power sharing, and power limiting, while operating with a variable voltage DC bus. This disclosure presents control techniques providing such features and performance comparisons between fixed DC bus systems and PV bus systems.

PV Bus Central Inverter Power Systems

[0031] As illustrated in FIG. 1, utility-scale solar projects include many PV panels **102a-102n** connected to a grid-tied central inverter unit **110** that is equipped with integrated Maximum Power Point Tracking (MPPT) functionality. Typically, multiple solar modules (e.g., 30 solar modules) are series-connected to form a PV string and PV strings (e.g., 10-24 PV strings) are parallel-connected at a combiner box (not shown). The combiner box is then connected to the central inverter unit **110**. The PV bus system **100** of this disclosure includes sets of PV panels **102a-102n**, energy storage devices **104a-104n**, and a grid-tied or islanding-capable central inverter unit **110**.

[0032] The sets of PV panels **102a-102n** are connected to common fixed DC buses **108a-108n**, respectively. Unlike a fixed DC bus architecture, distributed MPPT controllers connected to the sets of PV panels **102a-102n** may not be utilized in the PV bus system. Instead, MPPT is achieved within either the central inverter unit **110** or the distributed DC/DC controllers **106a-106n**, each of which includes a DC/DC converter.

[0033] The energy storage devices **104a-104n**, which may be vanadium flow batteries or lithium-ion batteries, are connected to the common, fixed DC buses **108a-108n** via the battery controllers **106a-106n**, respectively. The grid-tied or islanding-capable central inverter unit **110** converts DC electricity to AC electricity. The central inverter unit **110** includes a DC/DC controller **116** implementing MPPT functionality. The DC/DC controller **116** may include a step-up or step-down DC/DC converter. The central inverter unit **110** also includes an inverter **118** connected to the output of the DC/DC controller **116**. Alternatively, the DC/DC controller **116** and the inverter **118** may be hybridized into a single unit.

PV Bus Control Methods

[0034] As performance and stability issues arise when multiple DC/DC controllers with MPPT functionality simultaneously attempt to track a maximum power point from a common PV bus, it is difficult to integrate energy storage systems into these systems at the bus level. In instances, DC/DC controllers with MPPT functionality relying on voltage perturbation methods, such as the Perturb and Observe and Incremental Conductance MPPT methods, may inadvertently track the maximum power point based on the voltage perturbations of neighboring DC/DC controllers with MPPT functionality, resulting in poor performance and/or system instability. To address this issue, several novel distributed multiagent maximum power tracking methods that can operate with low-bandwidth or no-bandwidth communications may be used with the PV bus system **100**.

Control Mode Hand-Off

[0035] Control mode handoff is a distributed MPPT method that designates one of the DC/DC controllers **106a-106n** as a PV bus agent with exclusive MPPT control privileges. The MPPT control privilege is “handed-off” to a neighboring DC/DC controller with MPPT functionality after a prespecified duration, e.g., 5-15 seconds, or if certain operating conditions are met, such as a hand off to the DC/DC controller **106a-106n** associated with the battery or battery bank **104a-104n** with the lowest state of charge. The DC/DC controllers not currently operating in MPPT mode operate instead with a fixed duty cycle. The duty cycle ratio

may be set at each DC/DC controller’s **106a-106n** last known MPPT setpoint. FIG. 2 is a flowchart illustrating an example of a method **200** for control mode handoff.

[0036] At block **201**, the PV bus system initiates startup. At block **202**, the method **200** includes determining at the current DC/DC controller whether MPPT control is enabled. If MPPT control is enabled, an MPPT mode timer, which may also be referred to as an MPPT mode handoff timer or MPPT mode transition timer, is started at block **204**. Then, at block **206**, the local voltage and current, e.g., the voltage and current at the output of the PV string, are measured and the maximum power point voltage seeking direction ΔV is calculated based on the measured local power perturbations.

[0037] At block **208**, the method **200** includes determining whether the power limit is enabled. If the power limit is enabled, the method includes determining whether the output power is greater than the power limit at block **210**. Block **210** may include determining the output power based on the measured local voltage and current. If the output power is greater than the power limit, MPPT is performed in the direction of negative ΔV at block **212**. If the output power is not greater than the power limit, MPPT is performed in the direction of positive ΔV (which is also referred to herein as reverse MPPT) at block **214**. If the power limit is determined not to be enabled at block **208**, MPPT is performed at block **212**.

[0038] After MPPT is performed at block **212** or after reverse MPPT is performed at block **214**, the method **200** includes determining whether the MPPT mode timer is greater than a time limit. If the MPPT mode timer is determined to be greater than a time limit, the MPPT control mode is handed off to the next unit at block **218**. If the MPPT mode timer is not greater than the time limit at block **216**, the method **200** returns to block **202** to determine whether MPPT control is enabled. If MPPT control is not enabled, the previous duty cycle at the current unit’s last known MPPT setpoint is held. Then, the method **200** repeats block **216**, determining whether the MPPT mode timer is greater than the time limit.

Sequential MPPT Control

[0039] Sequential MPPT control is a modification of control mode handoff in which each distributed energy resource (DER) unit is allowed a single voltage perturbation before passing the MPPT control privilege to the next DC/DC controller.

Adaptive ΔV MPPT Control

[0040] Adaptive ΔV MPPT control operates by scaling the magnitude of the voltage or power perturbation based on a local device metric, such as current device power output or battery state of charge. In this mode, no communications between DC/DC controllers may be needed.

Power Limiting Control

[0041] Power export curtailment may be achieved with by applying a small modification to the underlying MPPT method while using the control mode handoff and adaptive ΔV techniques. This modification involves reversing the direction of a MPPT controller duty cycle adjustment during the voltage perturbation step when power export exceeds set limits. Power limiting control applied to the Perturb and Observe and Incremental Conductance methods are similar.

Fixed DC Bus Power Systems

[0042] The fixed DC bus power system **300** utilizes a single, fixed DC bus **308** to transfer power among all connected devices. MPPT and voltage matching may be provided by string-level MPPT controllers **103a-103n** and a central inverter **310** with MPPT functionality. Multiple wiring runs to the central inverter **310** may be eliminated.

[0043] The fixed DC bus power system **300** includes sets of PV panels **102a-102n**, energy storage devices **104a-104n**, DC/DC controllers **306a-306n**, which include DC/DC converters, and a grid-tied and/or islanding-capable central inverter unit **310**. The sets of PV panels **102a-102n** are coupled to a common fixed DC bus **308** via distributed MPPT controllers **103a-103n**. Each MPPT controller **103a-103n** is coupled between a respective PV panel **102a-102n** and the common fixed DC bus **308**. The energy storage devices **104a-104n**, which may be vanadium flow or lithium-ion batteries, are connected to the common, fixed DC bus **308** via bidirectional DC/DC battery controllers **106a-106n**, respectively. The grid-tied and/or islanding-capable central inverter unit **310** converts DC electricity to AC electricity and is capable of operating with a fixed DC input.

Fixed DC Bus Control Methods

[0044] The availability of a common voltage standard across all devices simplifies control design for fixed DC bus power systems **300**. Power sharing, power export limiting, and battery charge/discharge can be achieved by operating individual devices, e.g., the MPPT controllers **103a-103n**, in voltage/power droop mode, current limiting mode, and current tracking modes, respectively. DC bus voltage stability may be ensured by limiting voltage tracking to a single device per fixed DC bus system by utilizing power/voltage droop control methods. Voltage regulation may be assigned to either the central inverter **315** for maximum utility power export with limited battery charge/discharge, or the battery system for curtailed power grid power export with maximum battery charging. Voltage regulation assignment may be achieved via the same low bandwidth communications pathways as used in PV bus control mode handoff. Table 1 illustrates PV MPPT, inverter, and battery control modes during various system operation modes for PV bus and fixed DC bus power systems.

TABLE 1

Example PV Bus and Fixed DC Bus Device Control Modes.						
PV Bus				Fixed DC Bus		
		Power Limiting			Power Limiting	
		Power Export	with Battery		Power Export	with Battery
	MPPT	Limiting	MPPT	MPPT	Limiting	MPPT
Central Inverter	MPPT ¹	Power Limiting	Power Limiting	Voltage Regulation	Power Tracking ²	Power Tracking
		MPPT ¹	MPPT ¹			

TABLE 1-continued

Example PV Bus and Fixed DC Bus Device Control Modes.						
PV Bus				Fixed DC Bus		
		Power Limiting			Power Limiting	
		Power Export	with Battery		Power Export	with Battery
	MPPT	Limiting	MPPT	MPPT	Limiting	MPPT
D _{MPPT} Battery	N/A	N/A	N/A	MPPT	MPPT	MPPT
	Power Limiting	Power Limiting	MPPT ¹	Power Tracking	Power Tracking ²	Voltage Regulation
	MPPT ¹	MPPT ¹				

¹Control Handoff Enabled.

²Voltage/Power Droop Enabled.

Simulation

[0045] The above control methods may be implemented and applied to 3:2:1 and 1.6:0.6:1 (PV:Battery:AC) PV bus and fixed DC bus power system models. FIGS. 4 and 5 show examples of models for the PV bus and fixed DC bus systems, respectively. In the models, solar data **402** is filtered by a low-pass filter **404** and random noise **406** is added to the filtered solar data via an addition block **408**. The processed solar data, e.g., the irradiance applied to the PV panels, is then applied to the input of a PV array model **415** via a saturation limit block **410**. A temperature is also applied to the input of the PV array model **415** via a temperature model block **412**.

[0046] In the PV bus system model of FIG. 4, the measurement block **432** measures the voltage output from the PV array model **415**. In the fixed DC bus system model of FIG. 5, the measurement block **432** measures the voltage output from the MPPT controller **525**, which is coupled to the outputs of the PV array model **415**. The measurement block **434** measures the voltage of the battery **422**, the measurement block **436** measures the voltage at the location of the combiner box, and the measurement block **438** measures the voltage at the inverter **424**. The PV bus system model also includes a display block **418** coupled to the measurement output of the PV array model **415**. The display block **418** may display one or more measurements relating to the PV array model **415**, e.g., voltage and current.

Results

[0047] Simulation results are presented below comparing PV bus power systems and fixed DC bus systems. The above PV bus control methods may be utilized to provide PV bus power system functionally similar to that inherent in the fixed DC bus power system.

MPPT Tracking Time

[0048] The time tracking settings of centralized and distributed MPPT may not have a significant impact on performance when applied to typical solar insolation profiles. A survey of various MPPT methods, shown in Table 2 below, suggests that the tracking time of current MPPT methods may be sufficient to account for typical temporal variations in insolation. Compared to the frequency of variations in solar as illustrated in FIG. 6 by the power spectral density of solar data, current MPPT tracking speeds are adequate to track the majority of temporal insolation variations and a time-accelerated direct normal irradiance (DNI) profile is used in all simulations.

TABLE 2

Example Tracking Time Comparison of Various MPPT Methods.	
Example Tracking Methods	Example Tracking Times (seconds)
Differential Particle Swarm Optimization	0.5
Fill Factor	1.96
Modified Fill Factor	1.2
Conventional Perturb and Observe	0.2
Drift-free Perturb and Observe	0.15
Hybrid Ant Colony Optimization/P&O	2.41
Particle Swarm Optimization	12.41
Ant Colony Optimization	9.47

PV Bus Control Mode Handoff Performance

[0049] PV bus MPPT tracking and power export limiting performance achieved with control mode handoff may be evaluated by simulating an accelerated-time insolation profile with a PV+battery PV bus system. A digital twin PV observer model with true MPPT may also be simulated to provide a performance baseline. Results for control mode handoff with MPPT are shown in FIG. 7. As shown in FIG. 7, the inverter power **702** and the battery power **704** increase from about 1000 W to about 5000 W and then decrease back to about 1000 W. The total power curve **706** tracks the MPPT power curve **708**. In this simulation, power sharing is unforced and is automatically determined by the ratio of the line impedances between the batteries and central inverter and the PV bus, but can be altered via the introduction of a power-voltage droop bias using, for example, the controller of FIG. 12, which incorporates a droop stage. The total energy extracted from the PV panel matches baseline performance.

[0050] Results using control mode handoff with MPPT and a 3000 W inverter power export limit are shown in FIG. 8. As shown in FIG. 8, the battery power **704** increases from about 1000 W to over 6000 W and then decreases down to the power export limit **801**; the inverter power **702** increases from about 1000 W, is limited to the power export limit **801**, and then decrease to 0 W; and the total power curve **706** tracks the MPPT power curve **708**. In this case, active PV bus devices operate in unforced power sharing mode, which is the system operating in static duty cycle mode and which may correspond to the state **220** of FIG. 2, similar to the scenario in FIG. 7. When a power export limit is reached, e.g., the power export limit **801** illustrated in FIG. 8, the central inverter switches to reverse MPPT control to maintain a fixed power output and the battery absorbs the excess PV power. Control mode handoff may automatically adjust the power sharing ratios to extract maximum PV energy while limiting central inverter power export.

[0051] In both scenarios, overall system MPPT is comparable to baseline performance and power export tracking performs as expected. Slight tracking overshoot/undershoot is shown, but this is attributed to the aggressive simulation time acceleration factor and large ΔV . Improved performance is expected with real-time conditions and smaller voltage perturbation steps.

Fixed DC Bus Performance

[0052] Fixed DC bus MPPT power sharing and MPPT power sharing with inverter power export limit results are shown in FIGS. 9 and 10. In the non-export limiting case

relating to the results illustrated in FIG. 9, an inverter-to-battery droop ratio of 2:1 is established to demonstrate controlled power sharing. As illustrated in FIG. 9, the battery power **704** increases from about 700 W to over 3000 W and then decreases back to about 700 W; the inverter power **702** increases from about 1300 W to over 6000 W and then decrease back to about 1300 W; and the total power curve **706** tracks the MPPT power curve **708**.

[0053] In the power limiting case relating to the results illustrated in FIG. 10, a 3000 W power export limit is imposed. As shown in FIG. 10, the battery power **704** increases from 0 W to over 6000 W and then decreases back to 0 W; the inverter power **702** primarily tracks the 3000 W power export limit; and the total power curve **706** tracks the MPPT power curve **708**. A hybrid power droop/power limiting controller, which allows seamless and automatic transition between droop control and power limiting, may be applied within any bus-connected component. In aspects, the hybrid power droop/power limiting controller may be incorporated into the central inverter units, the inverters, the MPPT controllers, and/or the DC/DC controllers of this disclosure. An example of a hybrid power droop/power limiting controller **1100** is illustrated in FIG. 11. The hybrid power droop/power limiting controller **1100** includes a proportional-integral (PI) control stage **1110** and a droop stage **1120**. Alternatively, the PI control stage **1110** may be replaced by a proportional-integral-derivative (PID) control stage. The derivative gain of the PID control stage may be set to zero.

[0054] The hybrid power droop/power limiting controller **1100** includes a voltage input terminal **1102** and a current input terminal **1104**, which are applied to a multiplication block **1111**, which determines the product of a voltage value applied to the voltage input terminal **1102** and a current value applied to the current input terminal **1104**, and outputs the resulting power value. The power value may be subtracted from a bias value output from a bias block **1112** using a subtraction block **1113**. The output from the subtraction block **1113** is applied to a voltage gain block **1114**. The output from the voltage gain block **1114** is applied to a digital proportional-integral (PI) block **1116**. The voltage input **1102** is also applied to a droop gain block **1121**.

[0055] The output from the droop gain block **1121** is then subtracted from the output of the digital proportional-integral block **1116** by subtraction block **1122**. The output from the subtraction block **1122** is added to an offset value, e.g., 1500, output from an offset block **1119** by an addition block **1123**. The voltage input **1102** is then subtracted from the output of the addition block **1123** by subtraction block **1124**, the output of which is applied to voltage gain block **1126**. The output from the voltage gain block **1126** is then applied to digital PI block **1128**, which applies a PI control to the output from the voltage gain block **1126**. The output of the digital PI block **1128** is then provided to a DC/DC converter via output terminal **1130**.

[0056] System level power export limiting may be implemented with the battery operating in an idle state until an inverter export limit is reached, at which point the hybrid controller automatically transitions the DC/DC controllers into voltage regulation mode thereby absorbing excess available PV power.

System Efficiency

[0057] Conductor and converter losses for PV bus and fixed DC bus system for a single daily insolation curve with varying Inverter:Battery power capacity ratios are assessed (PV capacity is set at the combined inverter and battery capacity). A table of examples of device and conductor resistance values are presented in Table 3. Switching and other parasitic DC/DC converter losses are omitted.

[0058] FIG. 12 shows an example of a DC/DC controller 1200, which may implement each of the DC/DC controllers 106a-106n of FIG. 1. Also, the DC/DC controller 1200 or a modified version of the DC/DC controller 1200 may implement the central inverter unit 110, the MPPT controllers 103a-103n of FIG. 3, or the central inverter 315 of FIG. 3. The DC/DC controller 1200 includes a wireless antenna 1205 and a communications interface 1217 configured to communicate with other devices of a power system, e.g., other DC/DC controllers in a PV bus system. The DC/DC controller 1200 also includes a microcontroller 1210 and a power converter 1220. The microcontroller 1210 includes a processor 1211, e.g., a central processing unit (CPU), random access memory (RAM) 1212 coupled to the processor 1211, program memory 1213 coupled to the processor 1211, a data interface 1215 coupled to the processor 1211, driver circuitry 1216 coupled to the processor 1211 and to the power converter 1220, and a communications interface 1217 coupled to the processor 1211.

[0059] The data interface 1215 may be coupled to sensors to receive measurement data regarding aspects of the power system. For example, the data interface 1215 may be coupled to a battery fuel gauge, which may measure and/or determine the voltage, current, state of charge, or other information relating to the status of the battery or bank of batteries to which the DC/DC controller 1200 is connected. The program memory may store instructions for an MPPT control method 1214, which may be loaded into RAM 1212 and executed by the processor 1211. In other aspects, the driver circuitry 1216 is incorporated into the power converter 1220 or resides in the DC/DC controller 1200 separate from the microcontroller 1210 and the power converter 1220.

TABLE 3

Example Converter and Conductor Resistance Values	
Example Device/Conductor	Example Average DC Resistance
DC Bus-1500 V	
Inverter	0.005 Ω
DC Bus MPPT Converter	0.005 Ω
Total PV Series Resistance	0.485 Ω
Battery DC/DC Converter	0.005 Ω
Battery to DC Bus Line Segment	0.00025 Ω
PV to DC Bus Line Segment	0.00025 Ω
Total DC Bus Line Segment	0.00525 Ω
PV Bus (1200-2000 V)	
Inverter	0.005 Ω
Total PV Series Resistance	0.97 Ω
Battery DC/DC Converter	0.005 Ω
Battery to PV Bus Line Segment	0.00025 Ω
PV to PV Bus Line Segment	0.00025 Ω
Total PV Bus Line Segment	0.00525 Ω

[0060] The power converter 1220 may include a DC/DC converter, such as a DC/DC buck converter, that steps down a DC voltage received from a PV string and outputs a lower DC voltage onto a fixed DC bus. In the case of the hybridized central inverter unit 110 of FIG. 1 or the central inverter 315 of FIG. 3, the power converter 1220 may include a DC/AC inverter that converts the DC voltage on the DC bus to AC voltage suitable for feeding the power grid. The DC/AC converter may convert the DC voltage at a DC bus to AC voltage with suitable frequency and voltage magnitude to feed the power grid. The power converter 1220 may be controlled by the microcontroller 1210 via the driver circuitry 1216 to provide both AC power regulation and maximum power extraction. As used herein, "AC power regulation" means regulating the active power or reactive power output from the power converter according to active power or active power command signals, and "maximum power extraction" means dynamically moving the operating point of the solar power source to its highest point on the power curve (which may change in response to environmental changes such as irradiation and temperature changes) to have maximum amount of power output from the PV panels.

[0061] In the case of the hybridized central inverter unit 110 or the central inverter 315, the microcontroller 1210 is configured to send control signals to the power converter 1220 to regulate the PV power output from the PV strings according to various command signals and feedback signals. In the case of a hybridized central inverter unit 110 or the central inverter 315, the microcontroller 1210 is configured to send control signals to the power converter 1220 to regulate the active power or reactive power output from the power converter 1220 according to various command signals and feedback signals. The power converter 1220 may include any type of converter topology such as a half bridge converter, a full bridge converter, or a push-pull converter. In the case of a hybridized central inverter unit 110 or the central inverter 315, the power converter 1220 may include any type of DC to AC converter topology such as a two-level converter or a three-level converter. The power converter 1220 may include semiconductor switching devices (not shown), including but not limited to, integrated gate commutated thyristors (IGCTs) and insulated gate bipolar transistors (IGBTs).

[0062] FIG. 13 depicts an example of a fixed DC bus system architecture 1300 according to another aspect of this disclosure. The fixed DC bus system architecture 1300 includes at least one PV panel or PV string 1302, flow batteries 1312, and lithium-ion batteries 1322 electrically coupled to a fixed DC bus 1301 via DC/DC controllers 1304, 1314, 1324, respectively. Each DC/DC controller 1304, 1314, 1324 includes MPPT functionality. The fixed DC bus system architecture 1300 also includes two or more inverters 1332a-1332n coupled to the fixed DC bus 1301. The inverters 1332a-1332n may be configured as modular inverters, which may be selectively added to or removed from the fixed DC bus system architecture 1300 so that the fixed DC bus system architecture 1300 can be expanded or contracted as needed.

[0063] It should be understood that various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and accompanying drawings. It should also be understood that, depending on the example, certain acts or events

of any of the processes or methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques). In addition, while certain aspects of this disclosure are described as being performed by a single module or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or modules associated with, for example, PV panels, PV strings, or batteries.

[0064] In one or more examples, the described techniques may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored as one or more instructions or code on a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include non-transitory computer-readable media, which corresponds to a tangible medium such as data storage media (e.g., RAM, read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), flash memory, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer).

[0065] Instructions may be executed by one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the term “processor” as used herein may refer to any of the foregoing structure or any other physical structure suitable for implementation of the described techniques. Also, the techniques could be fully implemented in one or more circuits or logic elements.

What is claimed is:

1. A method comprising:
 - at a current DC/DC controller, measuring a local voltage and current when a maximum power point tracking (MPPT) control is enabled; and
 - in response to determining that a power limit is enabled:
 - determining an output power based on the local voltage and current;
 - comparing the output power to a power threshold;
 - performing reverse MPPT when the output power is less than or equal to the power threshold; and
 - performing MPPT when the output power is greater than the power threshold.
2. The method of claim 1, further comprising:
 - starting an MPPT mode timer when the maximum power point tracking (MPPT) control is enabled; and
 - handing off MPPT control to a next DC/DC controller after the MPPT mode timer exceeds a predetermined period.
3. The method of claim 2, further comprising:
 - in response to determining that MPPT control is enabled at the next DC/DC controller:
 - applying a single voltage perturbation from an energy source connected to the next DC/DC controller; and
 - handing off MPPT control to a third DC/DC controller.
4. The method of claim 1, wherein in response to determining that MPPT control is not enabled, holding a previous duty cycle at the current DC/DC controller's last known MPPT setpoint.

5. The method of claim 1, wherein the power threshold is adjusted based on real-time system metrics, including solar irradiance and battery state of charge.

6. The method of claim 1, further comprising:

- in response to determining, at a second DC/DC controller, that MPPT control is enabled;
- measuring a second local voltage and second current;
- determining that the power limit is not enabled; and
- performing MPPT at the second DC/DC controller.

7. The method of claim 1, further comprising determining a maximum power point voltage seeking direction (MPPT ΔV) based on the local voltage.

8. A PV bus system, comprising:

- a central inverter unit;
- a first PV panel coupled to the central inverter unit via a first PV bus;
- a second PV panel coupled to the central inverter unit via a second PV bus;
- a first DC/DC controller coupled to the first PV bus;
- a second DC/DC controller coupled to the second PV bus;

 wherein the first DC/DC controller is configured to:

- measure a local voltage and current when a local maximum power point tracking (MPPT) control is enabled; and
- in response to determining that a power limit is enabled:
 - determine an output power based on the local voltage and current;
 - compare the output power to a power threshold;
 - perform reverse MPPT when the output power is less than or equal to the power threshold; and
 - perform MPPT when the output power is greater than the power threshold.

9. The PV bus system of claim 8, wherein the first DC/DC controller is further configured to:

- start an MPPT mode timer upon determining that the maximum power point tracking (MPPT) control is enabled; and
- hand off MPPT control to the second DC/DC controller after the MPPT mode timer exceeds a predetermined period.

10. The PV bus system of claim 9, wherein the second DC/DC controller is further configured to hold a previous duty cycle at the second DC/DC controller's last known MPPT setpoint when a local MPPT control is not enabled.

11. The PV bus system of claim 8, wherein the first DC/DC controller is configured to perform reverse MPPT by adjusting a duty cycle in a direction of a positive maximum power point voltage seeking direction (MPPT ΔV).

12. The PV bus system of claim 8, wherein the first DC/DC controller is configured to perform MPPT by adjusting a duty cycle in a direction of a negative maximum power point voltage seeking direction (MPPT ΔV).

13. The PV bus system of claim 8, wherein the first DC/DC controller comprises a PID control stage and a droop stage.

14. The DC bus system of claim 13, wherein the PID control stage includes:

- a first voltage gain block; and
- a first PI controller coupled to an output of the first voltage gain block.

15. The DC bus system of claim 14, wherein the droop stage includes:

a droop gain block configured to apply a droop gain to an input voltage to obtain a droop gain voltage; and
 a first subtraction block coupled to an output of the droop gain block and configured to subtract the droop gain voltage from an output of the first PI controller to obtain an adjusted PI controller output.

16. The DC bus system of claim **15**, wherein the droop stage further includes:

an addition block coupled to an output of the first subtraction block and configured to add the adjusted PI controller output to a reference power to obtain a further adjusted PI controller output; and
 a second subtraction block coupled to the addition block and configured to subtract the input voltage from the further adjusted PI controller output.

17. The DC bus system of claim **16**, wherein the droop stage further includes:

a second voltage gain block coupled to an output of the second subtraction block; and
 a second PI controller coupled to an output of the second voltage gain block.

18. A PV bus system, comprising:

a central inverter unit;
 a first PV panel coupled to the central inverter unit via a first PV bus;
 a second PV panel coupled to the central inverter unit via a second PV bus;

a first DC/DC controller coupled to the first PV bus; and
 a second DC/DC controller coupled to the second PV bus;
 wherein the first DC/DC controller is configured to:

in response to determining that a local MPPT control is enabled, perform a single voltage perturbation for an energy resource connected to the first DC/DC controller; and

handing off MPPT control mode to the second DC/DC controller.

19. The PV bus system of claim **18**, wherein the first DC/DC controller is further configured to hold a previous duty cycle at the current DC/DC controller's last known MPPT setpoint when the local MPPT control is not enabled.

20. The PV bus system of claim **18**, wherein the first DC/DC controller is further configured to:

Measure a local voltage and current;
 determine MPPT ΔV based on the measured local voltage;
 in response to determining that a power limit is enabled:
 determining an output power based on the local voltage and current;
 comparing the output power to the power limit;
 performing reverse MPPT when the output power is less than or equal to the power limit; and
 performing MPPT when the output power is greater than the power limit.

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